MILITARY SPECIFICATION

WRENCH, UNIVERSAL SOCKET, SPLINED, SQUARE DRIVE, HIGH STRENGTH, THIN WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the universal sockets described herein shall consist of this document and the latest issue of MIL-W-8982.

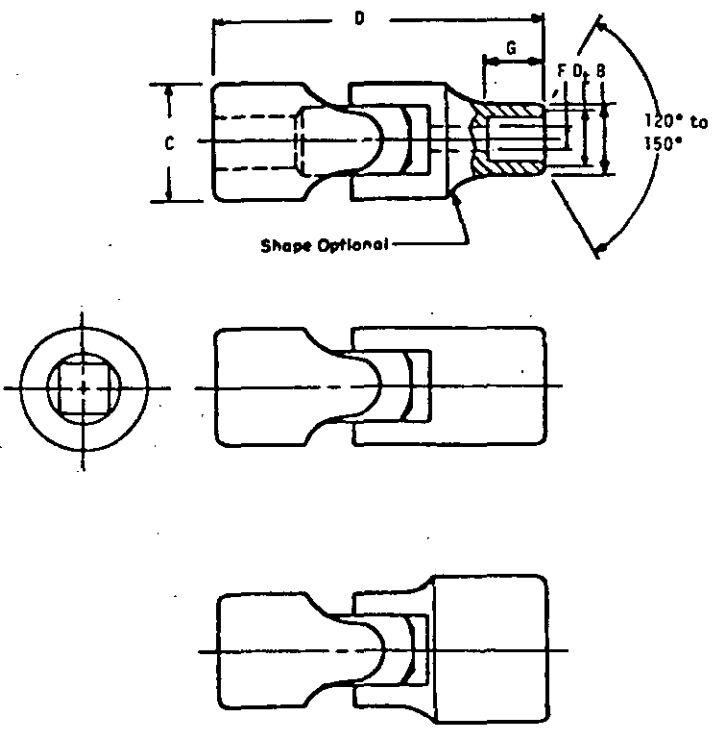


FIGURE 1. Universal sockets.

DESIGN REQUIREMENTS:

1. The universal socket shall conform to figure 1 and tables I, II, and III as applicable. Spline and D_t dimensions shall conform to MS33787. This wrenching design fits fasteners with the corresponding MS33787 spline as well as bihexagonal and hexagonal wrenching configurations.

AMSC N/A FSC 5120

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2. The female drive opening shall be designed for square drive dimensions in accordance with ANSI-B107.4. The drive opening shall be broached or punched in a smooth and well defined manner. The drive opening shall be firmly attachable to a corresponding male drive tang. Only the cross hole type shall be used.

3. All edges and corners capable of causing injury shall be rounded and chamfered with sharp edges removed. The inside edges of the spline opening shall be chamfered in accordance with MS33787.

4. A bolt clearance hole shall be provided at the base of the spline opening. The diameter of the bolt clearance hole shall conform to Tables I, II, and III.

5. The socket end shall be hardened to 48-52 HRC. The remaining components shall be hardened to 38-52 HRC.

6. The socket shall be chromium plated in accordance with QQ-C-320. Cadmium plating is not permitted.

PERFORMANCE REQUIREMENTS:

1. The universal socket shall withstand the endurance torque load and ultimate torque load values specified in tables I, II, and III. There shall be no damage or permanent deformation (set) to the wrench.

2. The endurance torque and ultimate torque loads shall be applied by any suitable manner or mechanical means employing in line load application. Loads shall be applied to the square drive test plug. The square drive plugs shall conform to the dimensions of ANSI B107.4 and be of suitable strength to withstand the ultimate strength of the socket. Measures shall be taken to prevent slippage of wrenches on mandrels during tests.

3. The socket shall be engaged with hexagon and spline mandrels in accordance with tables I, II, and III. It shall be subjected to the endurance torque load and ultimate torque load values specified in tables I, II and II. The load shall be held for for 10 seconds. The size of the hexagon mandrels in the engaged portion shall be equal to (+ 0.000, - 0.002) the basic size of the socket specified. Spline mandrel dimensions in the engaged portion shall be in accordance with MS33787, table I, except that C and W diamters shall be held to a minimum dimension (+ 0.002, - 0.000). Mandrels shall be hardened not less than 55 HRC and smoothly finished in the engagement area of wrench openings.

4. The socket shall be subjected to not less than 500 torque cycles on both the hexagon and spline mandrels in accordance with the endurance torque load and ultimate torque load values specified in tables I, II, and III. Engagement length shall be the same as that specified for the ultimate tests.

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TABLE I. Universal socket dimensions and test loads 1/4 square drive.

Dash No.	Basic Wr. Size	B Outside Dia. At Wr. End Max (in)	C Outside Dia. At Sq. End Max (in)	D Overall Length ± .060 (in)	F Bolt Clrc Hole Dia. Min. (in)	G Wrench Depth Min. (in)	Spline Test			Single Hex Test		
							Ult. Torque (in.-lb)	Endurance Torque (in.-lb)	L Engage Length (in)	Ult. Torque (in.-lb)	Endurance Torque (in.-lb)	L Engage Length (in)
06	.188	.310	.515	1.236	.160	.180	250	175	.091	150	100	.125
07	.219	.340	.515	1.236	.190	.200	300	210	.110	175	120	.125
08	.250	.375	.515	1.236	.220	.250	350	240	.125	250	175	.125
09	.281	.440	.515	1.299	.250	.280	550	385	.140	350	240	.141
10	.312	.460	.515	1.361	.280	.312	550	385	.156	450	315	.156
12	.375	.545	.515	1.486	.340	.343	550	385	.188	550	385	.188
14	.438	.619	.515	1.548	.410	.421	550	385	.219	550	385	.250

TABLE II. Universal socket dimensions and test loads 3/8 square drive.

Dash No.	Basic Wr. Size	B Outside Dia. At Wr. End Max (in)	C Outside Dia. At Sq. End Max (in)	D Overall Length ± .060 (in)	F Bolt Clrc Hole Dia. Min. (in)	G Wrench Depth Min. (in)	Spline Test			Single Hex Test		
							Ult. Torque (in.-lb)	Endurance Torque (in.-lb)	L Engage Length (in)	Ult. Torque (in.-lb)	Endurance Torque (in.-lb)	L Engage Length (in)
12	.375	.545	.750	1.970	.340	.343	1,000	700	.188	1,000	630	.188
14	.438	.635	.750	2.040	.410	.421	1,000	700	.219	1,000	700	.250
16	.500	.725	.750	2.080	.480	.437	1,000	700	.250	1,000	700	.313
18	.562	.825	.750	2.130	.530	.453	1,000	700	.281	1,000	700	.375
20	.625	.892	.750	2.190	.590	.531	1,000	700	.312	1,000	700	.438

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TABLE III. Universal socket dimensions and test loads 1/2 square drive.

Dash No.	Basic Wr. Size	B Outside Dia. At Wr. End Max (in)	C Outside Dia. At Sq. End Max (in)	D Overall Length ± .060 (in)	F Bolt Circ Hole Dia. Min. (in)	G Wrench Depth Min. (in)	Spline Test			Single Hex Test		
							Ult. Torque (in.-lb)	Endurance Torque (in.-lb)	L Engage Length (in)	Ult. Torque (in.-lb)	Endurance Torque (in.-lb)	L Engage Length (in)
16	.500	.725	1.125	2.812	.480	.437	1,750	1,225	.250	1,750	1,225	.313
18	.562	.825	1.125	2.890	.530	.453	1,750	1,225	.281	1,750	1,225	.375
20	.625	.892	1.125	2.950	.590	.531	1,750	1,225	.312	1,750	1,225	.438
22	.688	.972	1.125	3.020	.660	.562	1,750	1,225	.344	1,750	1,225	.500
24	.750	1.050	1.225	3.082	.720	.625	1,750	1,225	.375	1,750	1,225	.500
26	.812	1.160	1.225	3.140	.780	.656	1,750	1,225	.406	1,750	1,225	.562
28	.875	1.265	1.225	3.140	.840	.718	1,750	1,225	.438	1,750	1,225	.562
30	.938	1.375	1.225	3.360	.900	.781	1,750	1,225	.469	1,750	1,225	.625

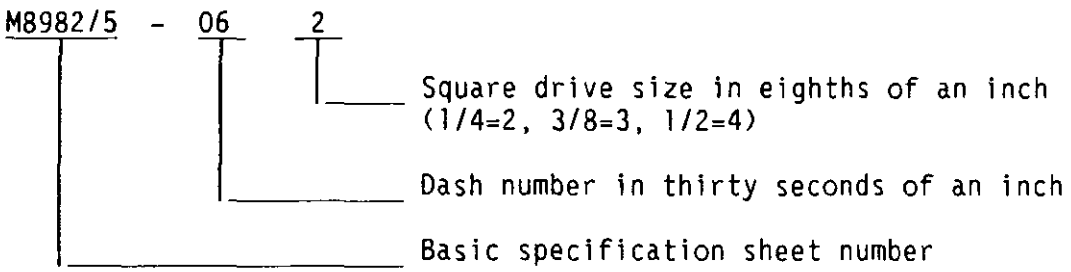
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NOTES:

1. Dimensions are in inches.

2. The socket shall be of uniform quality. It shall be free from nicks, splits, cracks, laps, seams and scale. It shall have no imperfections that will adversely affect serviceability, reliability or safety.

3. Example of a military part number:



Custodians:
Army - AV
Air Force - 11
Navy - AS

Review:
Army - GL, AR
Air Force - 99

Preparing Activity
Navy - AS

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